

Clinical Design Guidelines & Recommendations for ICU Sentinel

Prepared from Clinical Review Discussions with a Senior Consultant Intensivist

The consultant emphasized that ICU Sentinel should not focus solely on threshold-based alarms. The primary objective should be the early prediction of physiological deterioration before a patient reaches a critical state. The following recommendations were discussed for system design, signal analysis, patient prioritization, and clinical prediction logic.

1. General Clinical Philosophy

The consultant advised that most ICU emergencies do not occur suddenly. Physiological deterioration usually develops progressively and can often be identified through subtle changes in multiple vital signs before conventional alarm systems activate. The system should therefore focus on trend analysis rather than isolated threshold violations. Increasing heart rate, increasing respiratory rate, decreasing heart rate variability, declining perfusion, and worsening oxygenation should be considered collectively when estimating risk.

2. Tachycardia and Physiological Stress

The consultant explained that persistent tachycardia is frequently an early manifestation of infection, shock, blood loss, pain, fever, respiratory distress, or cardiovascular instability. Early warning pattern: • HR rises from baseline while BP remains normal. • HRV progressively decreases. • Respiratory rate begins to increase. • Temperature may rise if infection is present. Clinical concern increases when HR exceeds 110 bpm persistently and continues trending upward. The system should investigate whether tachycardia is associated with sepsis, shock, or respiratory failure rather than treating it as an isolated event.

3. Septic Shock Prediction

The consultant stated that septic shock is commonly preceded by several hours of physiological deterioration. Early indicators: • Rising temperature ($>38^{\circ}\text{C}$) • Increasing heart rate • Increasing respiratory rate • Decreasing HRV • Progressive multi-parameter instability Intermediate indicators: • MAP begins to fall. • Peripheral perfusion worsens. • Oxygen demand increases. Late indicators: • MAP < 65 mmHg • SBP < 90 mmHg • Multi-organ dysfunction The consultant recommended generating high-risk septic shock alerts several hours before hypotension develops.

4. Respiratory Failure Prediction

The consultant emphasized that respiratory rate is often the earliest indicator of respiratory deterioration. Observed progression: Stage 1: RR 18 $\rightarrow$ 22 $\rightarrow$ 26 breaths/min SpO₂ remains normal. Stage 2: RR exceeds 28 breaths/min. SpO₂ begins gradual decline. Stage 3: SpO₂ falls below 92%. Patient may require oxygen support. Stage 4: SpO₂ $< 90\%$. Acute respiratory failure becomes likely. The consultant recommended prioritizing respiratory rate trends over isolated oxygen saturation values.

5. Hypovolemic Shock

The consultant explained that internal bleeding and severe fluid loss often present with compensatory tachycardia before blood pressure falls. Expected progression: • HR increases. • Peripheral perfusion decreases. • Pulse pressure narrows. • MAP begins declining. • Shock

develops. The system should identify compensatory stages before overt hypotension occurs.

6. Cardiogenic Deterioration

The consultant advised that ECG abnormalities, worsening heart rate trends, reduced perfusion, and falling blood pressure may indicate cardiac instability. The system should monitor: • Sustained tachycardia • Arrhythmia patterns • Perfusion deterioration • Blood pressure trends These findings should contribute to a cardiac deterioration risk score.

7. Multi-Organ Dysfunction Risk

The consultant noted that severe deterioration frequently involves multiple physiological systems simultaneously. High-risk pattern: • HR increasing • RR increasing • Temperature increasing • BP decreasing • SpO2 decreasing The appearance of abnormalities across multiple systems should substantially increase patient priority.

8. Threshold Guidance

Suggested thresholds discussed during review: Heart Rate: Normal: 60–100 bpm Moderate concern: >100 bpm High concern: >110 bpm Critical concern: >130 bpm Respiratory Rate: Normal: 12–20 breaths/min Moderate concern: >22 High concern: >28 Critical concern: >30 SpO2: Normal: >95% Monitor closely: <94% High concern: <92% Critical concern: <90% MAP: Normal: >70 mmHg High concern: <65 mmHg Temperature: Fever concern: >38°C Severe concern: >39°C

9. False Alarm Reduction

The consultant stressed that sensor artifacts must not generate critical alerts. Potential causes: • Loose ECG electrodes • Motion artifacts • SpO2 probe displacement • Temporary communication failures The system should validate physiological plausibility before escalating alerts.

10. Patient Prioritization Strategy

The consultant recommended ranking patients according to: • Predicted deterioration probability • Estimated time-to-critical-event • Number of simultaneously worsening systems • Trend velocity Patients with rapidly worsening trends should be prioritized above patients with stable but abnormal values.

Final Clinical Recommendation

The consultant concluded that ICU sentinel should function as a proactive clinical decision-support system. The system should focus on early identification of shock, sepsis progression, respiratory failure, cardiac instability, and multi-organ deterioration through continuous analysis of ECG, SpO2, respiratory rate, blood pressure, temperature, and derived physiological features. Alerts should provide interpretable explanations and sufficient lead time for clinical intervention.